# Palette

Compression infrastructure for medical imaging, and the platform for AI that follows.

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# Background

The field of pathology is experiencing a transformation. Pathologists used to rely on physical slides for the examination of tissue. This required the pathologist and the slide to be co-located, often necessitating slide shipping and introducing delays into clinical workflows. It also limited the use of technologies like machine learning, which requires digital data as input and is rapidly becoming more prominent in both medical research and diagnostic workflows.

The introduction of whole-slide imaging devices allowed slides to be digitized, enabling pathologists to access slides anywhere and at any time: significantly streamlining clinical workflows. A limitation of these devices is the inherent manual workflow, where slides must be manually loaded into the device, a process that doesn't scale as the volume of slides to be digitized grows. For instance, medical institutions like Penn Medicine generate over a million slides annually; scanning these manually is simply not feasible.

High-throughput whole-slide imaging devices address this by automating the scanning process (Zarella & Rivera, 2022). Prominent hospitals, including Mayo Clinic, Ohio State, the University of Michigan, and Johns Hopkins, have dedicated digitization labs whose objective is to digitize physical slides. Efforts are also being made to digitize old repositories of archived slides to provide data for use in training machine learning models. For instance, Mayo Clinic has digitized over 11 million archival slides over the past two years, averaging more than 20,000 slides per day (Pramana, 2025). As of early 2025, Mayo Clinic's total digital archive exceeded 20 million whole-slide images (Mayo Clinic News Network, 2025).

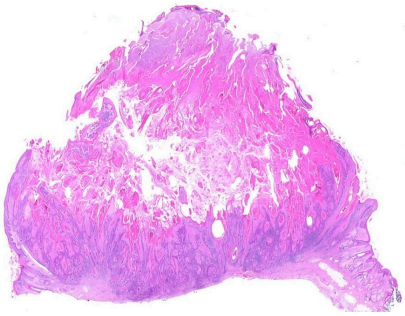

**Figure 1** (left): The digitization lab of Mayo Clinic. The lab consists of 29 scanners, each holding 450 slides. Employees review the digitized slides as they are processed before they are sent to storage.[1] **Figure 1** (right): A whole-slide image of tissue.

The output of a scan is a whole-slide image (WSI), which given the inherent detail of the scan typically ranges from 2 GB to over 10 GB. In practice, only portions of the WSI are retrieved at a time, requiring preprocessing prior to storage. This includes tiling the WSI into subsections, usually of size 512×512, before storing them in a pyramidal structure within the institution's Picture Archiving and Communication

[1] The most efficient scanners can generate one whole-slide image in roughly 60 seconds (Market Reports World, 2026).

System (PACS).[2] The size of these files, combined with growing digitization volumes, creates a storage challenge that is becoming one of the central barriers to broader adoption of digital pathology.

# Problem

A byproduct of this rapid digitization is the need for massive amounts of digital storage. Medical institutions such as Penn Medicine, scanning over 1 million slides per year, can generate on the order of a petabyte of WSI data annually, before accounting for redundancy, backups, and disaster-recovery copies (Ashman et al., 2022). This is not a one-time storage event; it becomes a growing liability because pathology operates on long retention horizons, and U.S. regulations require slides to be retained for at least 10 years (42 C.F.R. § 493.1105). As digitization expands to include both new clinical volume and historical archives, storage and data management are set to become dominant cost drivers, and in large real-world deployments storage has been reported as the single largest cost category (Ardon et al., 2023).

Using the case of Penn Medicine and assuming a conservative average WSI size of 3 GB, toward the lower end of the typical 2–10 GB range, 1 million slides per year produces approximately 3 PB of new WSI data annually. With a 10-year retention requirement, the stored archive grows linearly to ~30 PB before plateauing. Using AWS S3 Standard storage pricing of $0.021 per GB-month, annual storage cost scales proportionally with stored volume, reaching a steady-state rate of approximately $7.56M/year by year 10 and a cumulative spend of ~$37.8M over the first decade, excluding redundancy/backup multipliers and other cloud charges.[3] This is for a single department within a single institution.

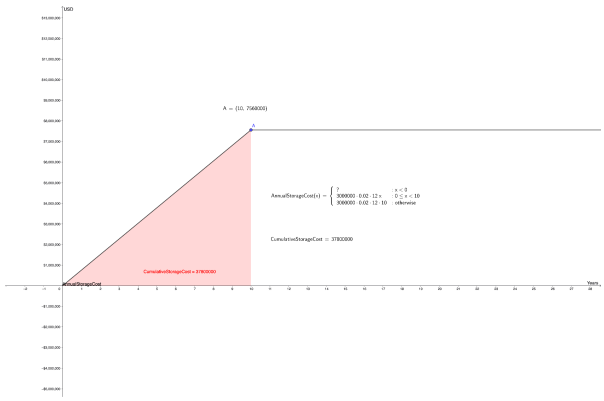


**Figure 2:** Annual storage cost for Penn Medicine under the conservative 3 PB/year scenario, assuming a 10-year slide retention requirement and AWS S3 Standard pricing at the ≥500 TB tier. Annual cost grows linearly to ~$7.56M/year at year 10 as the archive reaches 30 PB, then plateaus. The shaded area represents cumulative storage spend over the first decade (~$37.8M).

While the upfront cost of the scanner is often cited as the primary constraint on adopting whole-slide imaging, the ongoing cost of storage and infrastructure is equally prohibitive, and unlike a one-time hardware purchase, it grows every year (Market Reports World, 2026). Dr. Mark Zarella, Vice Chair of Digital and Computational Pathology at Penn Medicine and formerly of Mayo Clinic and Johns Hopkins Medicine, notes that institutions evaluating whether to adopt digital pathology frequently cite storage costs as a deciding factor, and in many cases it is the reason they choose not to digitize at all. The consequence is that a technology with demonstrated potential for improving diagnostics, research and powering adoption of AI in healthcare is being held back.

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[2] Pyramidal storage stores the WSI at multiple resolutions, with level 0 being the full-resolution image and increasing levels at progressively lower resolution. This allows for viewing the WSI without having to load the entire gigabyte-sized file into memory.

[3] This estimate uses AWS S3 Standard pricing at the ≥500 TB tier ($0.021/GB-month). In practice, hospitals vary in whether they store on-premises or in the cloud. On-premises storage carries a larger upfront fixed cost with lower recurring variable costs, while cloud storage shifts the burden to larger recurring variable costs. The core dynamic, that storage costs scale with archive size, applies in both cases.

# Product

Our product is a headless compression middleware layer installed between the whole-slide scanner and the institution's existing storage. The service ingests each newly scanned slide, converts scanner-native output into a standards-aligned pyramidal, tiled representation, and encodes the tiles using a compression algorithm developed to be visually lossless for diagnostic use while preserving fast random tile access for downstream clinical viewing and computational workflows. Importantly, the middleware is designed to integrate with existing operations: it does not require modifications to the scanner, does not replace the image management or viewer stack, and writes back into the institution's current storage environment using familiar file and metadata conventions. The product would not require FDA approval.

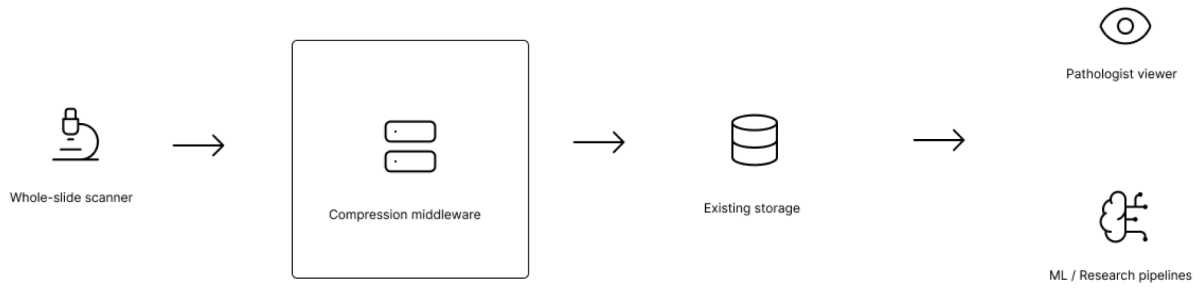


**Figure 3:** Architecture of the compression middleware. The physical slide is scanned using a whole-slide scanner, producing a WSI. Our middleware ingests the WSI after metadata hydration and compresses it before writing to the institution's existing storage (PACS) in a tiled-pyramidal format, from which pathologist viewers and ML/research pipelines read as before.

Compression is well documented in the computational pathology literature as an effective way to reduce WSI size while maintaining utility for clinical interpretation and research (Ashman et al., 2022). The practical question, therefore, is not whether compression works, but why it has not translated from research into deployment in clinical pipelines. The primary barrier is that pathologists lack the engineering capacity to build and maintain a production-grade compression pipeline. Their expertise is in diagnostics and research, not engineering. The tools that exist today are ad hoc scripts and internal workarounds, not deployable products a lab or hospital can install and operate.

In our preliminary tests, our compression pipeline reduced WSIs from an original size of 2–5 GB each to an average of 75 MB, a reduction ranging from approximately 25x to 65x depending on the original file size. This at a visually lossless quality level that preserves diagnostic and research utility.

The product will be installed and maintained by our team, and if additional compute is required, we will provide the necessary hardware or deployment support (see *Business Model* for details). This aligns with how hospitals typically procure specialized software from external vendors; Penn Medicine's Digital Pathology Lab alone maintains over eight hundred vendor software subscriptions.

# Market

## Overview

The global digital pathology market size was estimated at USD 1.47 billion in 2026 and is projected to reach USD 5.75 billion by 2034, growing at a CAGR of 18.65% (Fortune Business Insights, 2026). The software segment led the market with the largest revenue share in 2025 and is expected to grow at the fastest rate through 2033. North America accounted for the largest regional share at approximately 40% of global revenue (Grand View Research, 2026).

## Customer

We target academic research labs and hospitals as our core customer groups. The academic research lab segment was the largest by revenue in 2025, representing 45.76% of the market, while hospitals accounted for 37.15% as the second-largest segment (Grand View Research, 2026).

According to Dr. Zarella, any institution operating a whole-slide imaging scanner is a potential customer. As of 2024, over 4,000 such scanners were in operational use globally across research labs and hospitals, with more than 60% of pathology labs in North America and Europe now relying on whole-slide scanning systems for diagnostics and research. We aim to initially target the United States, which leads the market with over 1,500 scanners installed across our target customer groups; a natural starting point given our existing relationship with Penn Medicine. Geographic expansion into Canada, with roughly 900 scanners installed, follows naturally (Market Reports World, 2026).

## Competition

Our research indicates no existing product comparable to our offering, a headless compression middleware layer, something verified by Dr. Zarella and his colleague Dr. Mathew Francis, who previously held a position as a researcher at the Mayo Clinic. Dr. Zarella has expressed strong interest in deploying it within the Department of Pathology and Laboratory Medicine at Penn Medicine once developed. A similar emphasis was echoed by Dr. David Asch, Senior Vice President for Strategic Initiatives and Executive Director of the Penn Medicine Center for Health Care Innovation, who described the problem we are addressing as important. That said, there are adjacent players who address storage costs through different approaches.

The first category is lifecycle management vendors, which reduce cost by moving data to cheaper storage tiers over time rather than reducing file sizes. *Sectra Medical*, *Tiger Health Technology*, and *Glencoe* fall into this category. The second category is digital pathology image management systems (IMS), which provide end-to-end image management with some storage management functionality, though storage optimization is not their core focus. *PathAI*, *Proscia*, and *Philips* fall into this category.

These vendors are largely complementary to our offering; a customer could use our compression middleware alongside *PathAI* or *Proscia* for downstream analysis. Some scanners do apply compression at capture, but at significantly lower ratios than what our pipeline achieves at equivalent quality. Moreover, the most widely used whole-slide scanner globally, the *Leica Aperio GT 450*, outputs in uncompressed format, as does the third most popular scanner in the US, which is used at Penn Medicine.

# Business Model

We sell an enterprise on-premises subscription, aligned with how hospitals procure software from external vendors. Revenue consists of an annual site subscription, a one-time implementation fee, and a per-WSI usage fee that scales with throughput. Continuing our case of Penn Medicine at approximately 1 million WSIs per year and assuming a conservative 25× average reduction in stored bytes, the lower end of our demonstrated 25–65× compression range, storage drops from ~30 PB to ~1.2 PB. Using the same S3 Standard storage pricing of $0.021/GB-month, annual steady-state storage cost drops to ~$302k, compared to the original $7.56M. With our fees included, total annual spend is ~$642k, a ~12x decrease. A detailed pricing breakdown of our services is available in the Appendix.

# Team

I am a senior in the Management & Technology Program at the University of Pennsylvania majoring in Computer Science (BSE) in the School of Engineering, with a minor in Mathematics. Through Wharton, I pursue a second degree in economics with a dual concentration in Finance and Statistics. I am also part of Wharton's honors program; the Joseph Wharton Scholars program. I am originally from Norway and will be working on this project full-time in the US after graduation in May, 2026. More information about me, my projects, and research can be found on my website:
https://didrikwiigandersen.github.io/didrikwiigandersen/

# Goals

The short-term goal is for our product to be used by every academic research lab and hospital relying on whole-slide imaging, reducing infrastructure costs enough to stimulate broader adoption of digital pathology. We expect to deploy at Penn Medicine's Digital Pathology Lab by Q2 2026, sign two additional institutions by the end of Q3 and similarly two more by the end of the year. We begin development of *Palette Studio* in 2027. Long-term, I intend to grow Palette into the foundational infrastructure layer for medical imaging, expanding beyond compression into the platform capabilities described in the *Roadmap* section below. This will allow us to tap into a much larger market.

# Roadmap

We are currently developing the compression middleware. Dr. Zarella has agreed to host a deployment at Penn Medicine's Digital Pathology Lab for testing once development is complete.[4] Upon successful validation, we will offer the product for commercial sale.

## Phase 2: Expanding into other industries

A natural expansion is into biopharma. The use of whole-slide scanners in biopharma companies is increasing, with over 400 biopharma labs integrating them in drug development studies in 2023, and over $150 million allocated that year toward upgrading whole-slide scanners for preclinical pathology (Market Reports World, 2026). An example of a customer segment is companies conducting GLP histopathology studies. They generate enormous volumes of WSIs, and regulations require GLP study data to be retained for the life of the drug plus additional years. Contract research organizations like *Charles River*, *Covance*, and *INOTIV* are potential customers. The structural fit is identical: the same scanners, the same file

[4] We expect deployment by the end of March, 2026.

formats, and the same storage problem. Beyond biopharma, geospatial imaging presents a longer-term opportunity, with operators capturing petabytes of high-resolution images annually.

## Phase 3: Becoming a platform company

AI is rapidly entering diagnostic workflows in pathology and pharmaceutical research, but the infrastructure to support it does not exist. Despite over 1,200 AI-related publications in pathology and laboratory medicine between 2019 and 2024, only 22 AI-enabled devices received FDA clearance in that period, compared to over 770 in radiology (Garcia et al., 2025). The gap is in infrastructure. Training a clinical AI model today requires researchers to locate relevant data across fragmented, poorly indexed archives and manually construct datasets, a process that can take months. Once deployed, there is no standard infrastructure for monitoring data or model drift, a critical gap in an industry where diagnostic accuracy directly affects patient outcomes. As noted by Dr. Zarella, AI in clinical practice requires innovation to better characterize model performance, robustness, uncertainty, and risk.

Because our compression middleware processes every tile at ingest, we can extract feature embeddings per tile at near-zero marginal cost. This creates a tile-level feature index over an institution's entire slide archive, built passively as a byproduct of compression. No other company has this position: we sit at the single point through which all imaging data flows before it reaches storage, giving us compute access to every tile at the moment it enters the institution's system. This index unlocks a platform we call *Palette Studio*, with three layers:

- *Search and dataset creation.* Researchers today resort to randomized tile selection when constructing large datasets because there is no way to query archived slides by content. The alternative is manual screening, which is tedious, expensive, and time-intensive. With a tile-level feature index, a researcher could query "all tiles with necrotic tissue from 2024 colon biopsies" and receive results in seconds.[5] Dataset construction that currently takes weeks or months becomes near-instantaneous. This enables more precise, task-specific models. For instance, a classifier trained exclusively on necrotic tissue rather than a generic dataset. Since datasets can be generated faster, we foresee more such models being trained, with consequently higher accuracy and better patient outcomes, stimulating further adoption of AI in clinical workflows.
- *Pipeline development.* Once a dataset is created, researchers need to train, validate, and test models against it. *Palette Studio* provides the environment to do this without moving data off-platform, eliminating the transfer bottlenecks that plague current workflows. As emphasized by Dr. Mathew Francis, who develops software for whole-slide imaging workflows at Penn Medicine and previously at the Mayo Clinic, the size of these files makes transferring data between systems one of the most persistent friction points in computational pathology research. With *Palette Studio*, the dataset, the compute, and the model development live in one place, shareable across researchers within the organization.
- *Model monitoring and drift detection.* In a chronological evaluation of a Ki67 scoring model, model performance dropped significantly when trained on 2012–2018 data and tested on post-2018 cases, driven solely by changes in staining characteristics over time (Molnar et al., 2024). A similar concern was raised by Dr. Zarella during a lecture on February 17, 2026, stressing the need to detect when a model has drifted. Garcia et al. identify these generalizability challenges as a key barrier to clinical AI adoption, noting that model performance frequently degrades across sites and over time. Yet traditional quality management frameworks used in clinical laboratories were not designed for the rapidly iterative nature of AI systems that continue to evolve as they process new data (Garcia et al., 2025). There is currently no standard

[5] Every WSI is hydrated with metadata from the scanner before entering our pipeline. This metadata is incorporated into the feature embeddings for more precise tile search and retrieval.

infrastructure for detecting this drift in deployed pathology models. Because we continuously index incoming data, we can compare the feature distribution of new slides against a model's training distribution and flag when inputs have shifted beyond acceptable bounds. This enables detection of when model retraining is needed, improving model accuracy and diagnostic precision over time.

The strategic logic is straightforward: compression gets us installed, the index is built as a byproduct, and the platform monetizes the index. Every institution running our compression layer automatically generates the foundation for *Palette Studio*, an end-to-end platform for building, deploying, and maintaining machine learning models across healthcare and biopharma, with no additional integration required.

# Appendix

## Business Model

The middleware software license, including updates and a standard support package, is sold as an annual site subscription of $200,000. We also charge a one-time implementation fee of $20,000, which covers installation, integration testing, and a validation period to ensure the system operates correctly within the client's existing workflow.

Pricing scales with throughput. In addition to the site subscription, we charge $0.1 per WSI processed, billed periodically based on measured volume. The base annual subscription tier covers up to 300,000 WSIs/year; high-throughput sites can upgrade to tiers with higher included volumes and corresponding subscription pricing. Tiered pricing is summarized in the table titled *Usage Tiers*.

By default, we deploy the software onto the hospital's existing compute infrastructure (on-prem server). If this is infeasible, we offer required hardware for an additional $25,000 upfront.

Continuing our case of Penn Medicine digitizing ~1,000,000 WSIs/year, a conservative pricing estimate, without upgraded support, at the Tier 2 usage level, and without a hardware implementation, yields $340k/year in ARR. Specifically, an $220k/year site subscription (300k–1M tier) plus $100k/year in usage fees ($0.1/WSI), with a $20k one-time implementation fee in year one. Assuming a conservative 25× average reduction in stored bytes, the lower end of our demonstrated 25–65× compression range, storage drops from ~30 PB to ~1.2 PB. Using the same S3 Standard storage pricing of $0.021/GB-month, annual steady-state storage cost drops to ~$302k, compared to the original $7.56M. With our fees included, total annual spend is ~$642k, a ~12× decrease (excluding redundancy/backup multipliers).

**Annual Subscription**

| **Offering** | **What it includes** | **Price** |
|---|---|---|
| Site subscription (annual) | Software license, updates, and standard support package | $ 200,000 / year |
| Site subscription (annual) + hardware | Software license, updates, standard support package, and a server | $ 200,000 / year + $25,000 fixed up-front payment at installation |

**Implementation**

| **Offering** | **What it includes** | **Price** |
|---|---|---|
| Implementation (one-time) | Install, integration, and validation | $20,000 |

**Usage**

| **Offering** | **Cost driver** | **Price** |
|---|---|---|

| | | |
|---|---|---|
| Usage fee | WSIs processed | $0.1 / WSI |

**Usage Tiers**

| **Offering** | **Number of WSIs** | **Price** |
|---|---|---|
| Tier 1 | 0 - 300,000 WSIs / annually | Included with subscription |
| Tier 2 | 300,000 - 1,000,000 WSIs / annually | $20,000 added to the annual site subscription. |
| Tier 3 | Above 1,000,000 WSIs / annually | $50,000 added to the annual site subscription. |

**Support Tiers**

| **Offering** | **Response time** | **Price** |
|---|---|---|
| Tier 1 | 2 business days | Included with subscription |
| Tier 2 | 1 business day | + 10% of annual subscription is added per year. |
| Tier 3 | 4 business hours | + 20% of annual subscription is added per year. |

# References

Ardon O, Klein E, Manzo A, Corsale L, England C, Mazzella A, Geneslaw L, Philip J, Ntiamoah P, Wright J, Sirintrapun SJ, Lin O, Elenitoba-Johnson K, Reuter VE, Hameed MR, Hanna MG. Digital pathology operations at a tertiary cancer center: Infrastructure requirements and operational cost. J Pathol Inform. 2023 May 16;14:100318. doi: 10.1016/j.jpi.2023.100318. PMID: 37811334; PMCID: PMC10550754.

Ashman K, Zhuge H, Shanley E, Fox S, Halat S, Sholl A, Summa B, Brown JQ. Whole slide image data utilization informed by digital diagnosis patterns. J Pathol Inform. 2022 Jun 30;13:100113. doi: 10.1016/j.jpi.2022.100113. PMID: 36268057; PMCID: PMC9577055.

Fischer M, Neher P, Schüffler P, Ziegler S, Xiao S, Peretzke R, Clunie D, Ulrich C, Baumgartner M, Muckenhuber A, Dias Almeida S, Gőtz M, Kleesiek J, Nolden M, Braren R, Maier-Hein K. Unlocking the Potential of Digital Pathology: Novel Baselines for Compression. J Pathol Inform. 2025;17:100421. doi: 10.1016/j.jpi.2025.100421.

Fortune Business Insights. "Digital Pathology Market Size, Share & Industry Analysis Report." Fortune Business Insights. Accessed February 21, 2026.

Garcia CA, Reed KA, Lantz E, Day P, Zarella MD, Hart SN, Will E, Skiffington JG, Rice M, Novak DA, McClintock DS. Establishing a comprehensive artificial intelligence lifecycle framework for laboratory medicine and pathology: A series introduction. Am J Clin Pathol. 2025 Sep 9;164(3):424-437. doi: 10.1093/ajcp/aqaf069. PMID: 40650598.

Grand View Research. "Digital Pathology Market (2026–2033): Size, Share & Trends Analysis Report." Grand View Research (Report ID: 978-1-68038-386-7). Accessed February 21, 2026.

Market Reports World. "Research Slide Scanner Market Size, Share, Growth, and Industry Analysis, By Type (Brightfield, Fluorescence), By Application (University, Pharmaceutical Company, Hospital, Others), Regional Insights and Forecast to 2033." Market Reports World. Last updated February 2, 2026. Accessed February 21, 2026.

Mayo Clinic News Network. "Mayo Clinic launches Mayo Clinic Digital Pathology to modernize pathology, speed medical breakthroughs." January 13, 2025. Accessed February 22, 2026.

Molnar C, Tavolara TE, Garcia CA, McClintock DS, Zarella MD, Han W. IHCScoreGAN: An unsupervised generative adversarial network for end-to-end ki67 scoring for clinical breast cancer diagnosis. Proceedings of The 7th International Conference on Medical Imaging with Deep Learning, in Proceedings of Machine Learning Research. 2024;250:1011-1025. Available from https://proceedings.mlr.press/v250/molnar24a.html.

Office of the Federal Register. "42 CFR § 493.1105 — Standard: Retention requirements." Electronic Code of Federal Regulations. Current through February 19, 2026. Accessed February 21, 2026.

Pramana, Inc. "How Pramana Helped Mayo Clinic Digitize 11M Pathology Slides in Just Two Years." TissuePathology.com, March 20, 2025. Accessed February 22, 2026.

Zarella MD, McClintock DS, Batra H, Gullapalli RR, Valante M, Tan VO, Dayal S, Oh KS, Lara H, Garcia CA, Abels E. Artificial intelligence and digital pathology: clinical promise and deployment

considerations. J Med Imaging (Bellingham). 2023 Sep;10(5):051802. doi: 10.1117/1.JMI.10.5.051802. Epub 2023 Jul 31. PMID: 37528811; PMCID: PMC10389766.

Zarella MD, Rivera Alvarez K. High-throughput whole-slide scanning to enable large-scale data repository building. J Pathol. 2022 Jul;257(4):383-390. doi: 10.1002/path.5923. Epub 2022 Jun 8. PMID: 35511469; PMCID: PMC9327504.